

Microbial Sump to Soil Cultivator (MSSC)

Provisional Patent Summary (Draft for Counsel Review)

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Field of the Invention

Agricultural microbiology; integrated pond/bog water treatment, microbial cultivation, and sump to soil irrigation systems that generate compost tea byproduct while conditioning irrigation with beneficial consortia.

Background

Conventional farm ponds and bog filters are utilized primarily for water clarity, not as tunable bioreactors. Compost tea is often brewed separately, disconnected from irrigation loops. There is a need for a closed loop system that recirculates runoff, cultivates beneficial microbes in situ, and outputs inoculated irrigation water with optional concentrate capture.

Summary of the Invention

The MSSC routes runoff and pond water through a staged bog reactor and sump under controlled aeration and nutrient dosing, cultivating target microbial communities. A portion of the recirculated flow is conditioned and delivered to irrigation infrastructure; a bleed stream is captured as compost tea concentrate. Sensors (DO/ORP/pH/EC/temperature) and optional AI guidance adjust operating points over crop stages.

Representative Embodiments

A representative installation includes: (i) a ~60ft diameter pond; (ii) a 10×40ft bog filter with graded media and plantings; (iii) a sump with recirculation pump, aeration manifold, and C/N dosing ports; (iv) a tee to irrigation (drip/spray) and a bleed line to a compost tea tank.

Claims (Illustrative)

1. A sump centered microbial cultivation system comprising: a pond, a planted biofilter, and a sump configured to recirculate water under controlled aeration and nutrient dosing, wherein the system outputs irrigation water inoculated with beneficial microbial consortia.
2. The system of claim 1, wherein a bleed stream is diverted to a vessel to produce a compost tea concentrate as a byproduct of recirculation.
3. The system of claim 1, wherein the planted biofilter comprises staged gravel/media beds with redox zoned microbial habitats.
4. The system of claim 1, further comprising sensors to measure at least one of dissolved oxygen, oxidation reduction potential, electrical conductivity, pH, and temperature, and a controller configured to adjust aeration and dosing accordingly.
5. The system of claim 1, wherein rotational waterfowl grazing primes field rows prior to planting and is followed by application of the inoculated irrigation water.
6. The system of claim 1, wherein multi species manures and plant residues are incorporated into the bioreactor under conditions that mitigate pathogen risk while promoting nutrient cycling.
7. The system of claim 1, wherein AI assisted control adjusts microbial cultivation targets according to crop phenological stage.

Figures

Figure 1. Pond + Bog + Sump Layout (not to scale).

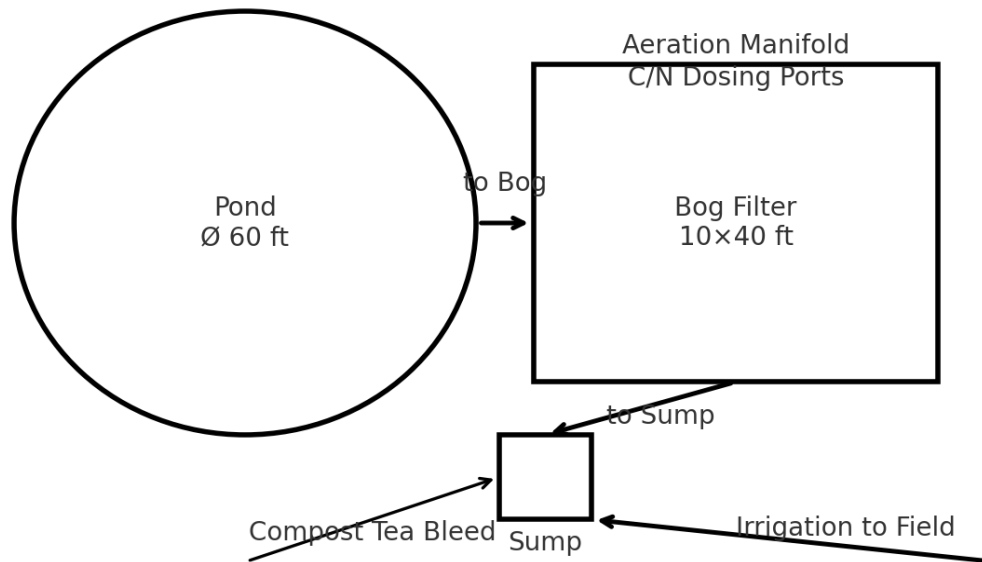


Figure 2. Process Flow Diagram.

